Modular Forms

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Preface

Today, September 1st 2026, marks my 25th birthday. Usually, resolutions are made around New Year's, but to me this seemed like an appropriate time as well. I have self-studied (pure) mathematics on the side before, but never thoroughly nor was it well-structured. This time I intend to change that and aim to study the lecture notes on modular forms by Dr P. Bruin and Dr S. Dahmen entirely and complete all the problems. In this document I will write my solutions and include thoughts I have whilst reading the text.
I am in no rush, so an additional challenge is that I will not use any form of AI in solving the problems. The reason is simply that I notice that it becomes too easy and I believe in the value of the struggle of not knowing how to tackle a problem. Of course, I will allow myself to use the more 'classic' sources such as Wikipedia, StackExchange and papers. For this reason, my solutions might contain errors.[1] This project really is a challenge to me and my patience. Knowing myself, I would love to learn all sorts of topics in pure mathematics, so preferably as fast as possible too. To that end, I force myself to only focus on Modular Forms and not use AI to solve problems quicker. I believe (or hope) that the result will be more rewarding than superficially studying several subjects at once in a short time. After all, this project is my own so who am I kidding in that case?
Anyhow, I'm rambling a bit. Why modular forms then? The honest answer is no particular reason other than there being some inexplicable attraction. One quickly reads about the importance of the subject in maths and theoretical physics. I have yet to discover that, but for now I am satisfied with the curiosity it ignited in me.

Structure

For each chapter I will start by stating my thoughts and questions I had whilst reading the main text. I will indicate this by ***Pensées***. These comments might be silly and hopefully I will look back at them later and be able to answer them. When I (think I) know the answer, still, I will leave the questions here and edits by future me will be indicated in red.
I will attempt to solve all the problems. I would like to stress that my solutions will be far from polished and aim to show my thinking. Therefore, in the solutions, too, there will be modifications in red making them possibly lengthy.

With the disclaimers out of the way, let's get started.

[1] For someone reading this document and spotting errors, please let me know: kianshams2001@gmail.com.

1 The modular group

Pensées

- It is explained that 'modular' refers to the moduli space of complex elliptic curves, but what is 'form' actually? I know of differential forms, bi-linear forms etc. and then how are these modular forms related to said moduli space? Hopefully the relationship with elliptic curves will be clear when reaching the corresponding chapter.

- I'm curious why $\mathbb{R}$-linear independence is necessary in the definition of a lattice. Although they do not explicitly define this, I think $\omega_1, \omega_2 \in \mathbb{C}$ are $\mathbb{R}$-linearly dependent if $\exists \lambda \in \mathbb{R}$ such that $\omega_1 = \lambda\omega_2$ (or vice versa). I think this is the 'full rank' part of a lattice in $\mathbb{C}$ in the definition, because else the lattice would be one-dimensional. If so, making this explicit would have been nice. Also because any lattice $\omega_1\mathbb{Z} + \omega_2\mathbb{Z} = \omega_2\Lambda_z$ for $z \in \mathbb{C} \setminus \mathbb{R}$ by the assumption of $\mathbb{R}$-linear independence. Wlog $z = \omega_1/\omega_2 \in \mathbb{H}$ so any homogeneous function $F : \mathcal{L} \longrightarrow \mathbb{C}$ is completely determined by its values on Λ_z so induces a function $f : \mathbb{H} \longrightarrow \mathbb{C}$ via $f = F(\Lambda_z)$ for $z \in \mathbb{H}$ Also, why then are we only concerned with full rank lattices?

- Cool! Lattices in $\mathbb{C}$ are in 1-to-1 with complex elliptic curves. Curious to learn how that works later. (Actually, I also know already (merely as a trivia fact) that complex elliptic curves are Riemann surfaces. For the example of a torus I know how a lattice corresponds to it. Still, the general case will become clear later I hope.)

- They say that one is naturally led to study a function $F : \mathcal{L} \longrightarrow \mathbb{C}$ satisfying $F(\lambda\Lambda) = F(\Lambda)$, $\forall \lambda \in \mathbb{C}^*$ (this is modulo homothety), but turns out to be too restrictive. Hopefully, it will be clear why that is the case. Not explained at this point. But accepting this fact, homogeneous functions of weight k are the next best alternative (?) which motivates the study of modular forms later (I secretly already know a tiny bit about it, e.g. the definition). Example of homogeneous function of weight k is the Eisenstein series, which I think is also a modular form and is relevant in physics (read: string theory) too.

1.1

(i) " $\Longleftarrow$ ": This direction we assume $\begin{pmatrix}\omega_1' \\ \omega_2'\end{pmatrix} = \gamma \begin{pmatrix}\omega_1 \\ \omega_2\end{pmatrix}$ for some $\gamma = \begin{pmatrix}a & b \\ c & d\end{pmatrix} \in \mathrm{GL}_2(\mathbb{R})$. Then $\omega_1' = a\omega_1 + b\omega_2$ and $\omega_2' = c\omega_1 + d\omega_2$, therefore $\Lambda' \subseteq \Lambda$.[2] Of course γ^{-1} exists so the converse also holds, i.e. $\Lambda' = \Lambda$.

" $\Longrightarrow$ ": $\Lambda' = \Lambda \implies \exists a, b, c, d \in \mathbb{Z}$ such that $\omega_1' = a\omega_1 + b\omega_2$ and $\omega_2' = c\omega_1 + d\omega_2$. Likewise $\exists \alpha, \beta, \delta, \theta \in \mathbb{Z}$ such that $\omega_1 = \alpha\omega_1' + \beta\omega_2'$ and $\omega_2 = \delta\omega_1' + \theta\omega_2'$. Plugging these relations in the first yields

$$\begin{aligned} & a(\alpha\omega_1' + \beta\omega_2') + b(\delta\omega_1' + \theta\omega_2') \overset{!}{=} \omega_1 \\ \Longrightarrow\ & a\beta + b\theta = 0 \ \wedge\ a\alpha + b\delta = 1, \quad \text{likewise} \\ & c(\alpha\omega_1' + \beta\omega_2') + d(\delta\omega_1' + \theta\omega_2') \overset{!}{=} \omega_2 \\ \Longrightarrow\ & c\alpha + d\delta = 0 \ \wedge\ c\beta + d\theta = 1. \end{aligned}$$

[2] Although the notation $\Lambda' \subseteq \Lambda$ is not introduced in the notes, I hope it is clear what I mean: all lattice points of Λ' are contained in Λ. This is different from homothety.

Compactly this is simply the requirement that

$$\underbrace{\begin{pmatrix} a & b \\ c & d \end{pmatrix}}_{\gamma} \begin{pmatrix} \alpha & \beta \\ \delta & \theta \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \tag{1}$$

In other words $\gamma \in \mathrm{GL}_2(\mathbb{Z})$.

(ii) At first glance I don't understand what proposition 1.2 has to do with this part as stated in the hint... By elimination: part (i) and (ii) probably aren't useful...
Ah ok I see now, because for all $\gamma \in \mathrm{GL}_2(\mathbb{Z})$ $\det(\gamma) = \pm 1$. In order for $z' = \omega_1'/\omega_2'$ to remain in $\mathbb{H}$ we must have $\mathrm{Im}(z') > 0$. By proposition 1.2 (i) we have $\mathrm{Im}(\gamma z) = \frac{\det(\gamma)\mathrm{Im}(z)}{|j(\gamma,z)|^2} = \mathrm{Im}(z')$.
So we must have $\det(\gamma) > 0$, meaning $\det(\gamma) \overset{!}{=} 1$ or in other words $\gamma \in \mathrm{SL}_2(\mathbb{Z})$.
To make this proof complete really is just a copy of part (ii) and including this minor observation.

1.2

Let $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $\gamma' = \begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix}$ be elements of $\mathrm{GL}_2(\mathbb{R})$. Then we simply work out

$$\begin{aligned} \gamma(\gamma' z) = \gamma\Big(\frac{a'z+b'}{c'z+d'}\Big) &= \frac{a\Big(\frac{a'z+b'}{c'z+d'}\Big) + b}{c\Big(\frac{a'z+b'}{c'z+d'}\Big) + d} \\ &= \frac{aa'z + ab' + bc'z + bd'}{ca'z + b'c + dc'z + dd'} \\ &= \frac{(aa'+bc')z + ab' + bd'}{(ca'+dc')z + b'c + dd'} =: (*). \end{aligned}$$

On the other hand,

$$\begin{aligned} (\gamma\gamma')\cdot z = \Big(\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix}\Big)\cdot z &= \begin{pmatrix} aa'+bc' & ab'+bd' \\ a'c+c'd & b'c+dd' \end{pmatrix}\cdot z \\ &= \frac{(aa'+bc')z + ab' + bd'}{(a'c+c'd)z + b'c + dd'} = (*). \text{ qed.} \end{aligned}$$

1.3

I am confused by the two parts of this question, because a *transitive action* of a group G on a set X means that for any $x, y \in X$ there exists a $g \in G$ such that $g \cdot x = y$. However, part b of this question then asks to show that for certain γ there are fixed points. How can the action be transitive and have a fixed point?
I am confusing two things here. If there is a γ for which there is an $x \in \mathbb{H}$ such that $\gamma \cdot x = x$ then there could still be a γ' such that $\gamma' \cdot x \neq x$ of course.

a. My initial idea was to brute force: compute $\gamma \cdot z = z'$ for $z, z' \in \mathbb{H}$ and show $\mathrm{Re}(\gamma \cdot z) = \mathrm{Re}(z')$, $\mathrm{Im}(\gamma \cdot z) = \mathrm{Im}(z')$ is always solvable, but this becomes messy.
There is a much cleaner idea which I owe to David C. Ullrich from the blogpost:

https://math.stackexchange.com/questions/2529664/how-to-show-the-action-of-sl-2-mathbb-r-on-the-complex-upper-half-plane-is-t . Namely, for any two points z, z' on the upper half plane first map e.g. point z' to i and then i to z. Because (**CLAIM**) we can map i to any point in the upper half plane by $\mathrm{SL}_2(\mathbb{R})$ and then we simply invert this and compose the two to find a map between the two points z, z'.

Let's prove **CLAIM**. First we need a map that sends $i \mapsto \lambda i$ for $\lambda \in \mathbb{R}_{>0}$ to fix the imaginary part and then $i \mapsto i + x$ for $x \in \mathbb{R}$ to fix the real part. The group elements that do this are $\begin{pmatrix} \sqrt{\lambda} & 0 \\ 0 & 1/\sqrt{\lambda} \end{pmatrix}$ and $\begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}$ which are both in $\mathrm{SL}_2(\mathbb{R})$. This proves $i \mapsto x + \lambda i$ upon composition.

Let's make it explicit that we can now use this to map any two points in the upper half plane with an $\mathrm{SL}_2(\mathbb{R})$ element:

Write $z' = \alpha + \beta i$, then $i \mapsto z'$ is implemented by

$$\begin{pmatrix} 1 & \alpha \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \sqrt{\beta} & 0 \\ 0 & 1/\sqrt{\beta} \end{pmatrix} = \begin{pmatrix} \sqrt{\beta} & \alpha/\sqrt{\beta} \\ 0 & 1/\sqrt{\beta} \end{pmatrix} \tag{2}$$

so $z' \mapsto i$ is implemented by $\begin{pmatrix} \sqrt{\beta} & \alpha/\sqrt{\beta} \\ 0 & 1/\sqrt{\beta} \end{pmatrix}^{-1} = \begin{pmatrix} 1/\sqrt{\beta} & -\alpha/\sqrt{\beta} \\ 0 & \sqrt{\beta} \end{pmatrix}$. Likewise write $z = \epsilon + \delta i$ then $i \mapsto z$ via

$$\begin{pmatrix} \sqrt{\delta} & \epsilon/\sqrt{\delta} \\ 0 & 1/\sqrt{\delta} \end{pmatrix} \tag{3}$$

and to check we have

$$\begin{pmatrix} \sqrt{\delta} & \epsilon/\sqrt{\delta} \\ 0 & 1/\sqrt{\delta} \end{pmatrix} \begin{pmatrix} 1/\sqrt{\beta} & -\alpha/\sqrt{\beta} \\ 0 & \sqrt{\beta} \end{pmatrix} \cdot z' \tag{4}$$

$$= \begin{pmatrix} \sqrt{\frac{\delta}{\beta}} & -\alpha\sqrt{\frac{\delta}{\beta}} + \epsilon\sqrt{\frac{\beta}{\delta}} \\ 0 & \sqrt{\frac{\beta}{\delta}} \end{pmatrix} \cdot z' \tag{5}$$

$$= \frac{\sqrt{\frac{\delta}{\beta}} z' - \alpha\sqrt{\frac{\delta}{\beta}} + \epsilon\sqrt{\frac{\beta}{\delta}}}{\sqrt{\frac{\beta}{\delta}}} \tag{6}$$

$$= \frac{\delta}{\beta}(\alpha + \beta i) - \alpha\frac{\delta}{\beta} + \epsilon \tag{7}$$

$$= \epsilon + \delta i = z. \tag{8}$$

Where $\begin{pmatrix} 1/\sqrt{\beta} & -\alpha/\sqrt{\beta} \\ 0 & \sqrt{\beta} \end{pmatrix} \in \mathrm{SL}_2(\mathbb{R})$ proving transitivity of the action on $\mathbb{H}$.

b. I immediatly note that $a + d$ is the trace of the matrix, but what does this have to do with the fixed points? (nothing immediate I think)

Next, I thought $\gamma \cdot z \overset{!}{=} z$ does not yield a relation including an $a + d$ term, but $\gamma^2 \cdot z \overset{!}{=} \gamma \cdot z$ does. Then I collected terms and naively imposed the coefficient of z^2 terms to match, but this led to an inconsistency. That in turn made me realise we need to be careful because z^2 may not be in $\mathbb{H}$

eventhough z is (e.g. $z \in \mathbb{H}$ for $\mathrm{Re}(z) \in \mathbb{R}_{<0}$). So, I went back to $\gamma \cdot z \overset{!}{=} z$ to write

$$\begin{aligned} \gamma \cdot z = \frac{az+b}{cz+d} &\overset{!}{=} z \\ \iff cz^2 + (d-a)z - b &= 0 \end{aligned}$$

Recalling secondary-school mathematics, the roots are

$$z = \frac{-(d-a) \pm \sqrt{(d-a)^2 - 4c(-b)}}{2c} \tag{9}$$

and realising that $z \in \mathbb{H}$ we must have $(d-a)^2 - 4c(-b) < 0$ and only the $+$ in $\pm$ is in the upper half plane (corresponding to there only being one fixed point). Writing out the condition

$$\begin{aligned} &(d-a)^2 - 4c(-b) = d^2 - 2ad + a^2 + 4bc \\ &= d^2 + a^2 + 2bc - 2 \qquad (\text{using } ad - bc = 1) \\ &= (a+d)^2 - 2ad + 2bc - 2 \\ &= (a+d)^2 - 4 < 0 \iff |a+d| < 2. \end{aligned}$$

(N.B. This solution excludes $\gamma = \pm\mathbf{1}$, as stated in the problem, because then $c = 0$ which causes problems in equation 9.)

1.4

a. Write $H \coloneqq \left\{ \begin{pmatrix} a & b \\ -b & a \end{pmatrix} \,\middle|\, a, b \in \mathbb{R},\ a^2 + b^2 = 1 \right\}$. Then clearly $H \subseteq K$, because

$$\begin{aligned} \begin{pmatrix} a & b \\ -b & a \end{pmatrix} \cdot i &= \frac{ai+b}{-bi+a} \\ &= \frac{(ai+b)(bi+a)}{a^2+b^2} \\ &= -ab + a^2 i + b^2 i + ab \\ &= (a^2+b^2)i = i. \end{aligned}$$

Conversely,

$$\begin{aligned} \begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot i = \frac{ai+b}{ci+d} &\overset{!}{=} i \\ \iff ai + b &= -c + di \\ \iff a = d \ \wedge\ b &= -c. \end{aligned}$$

Hence, $H = K$.

b. First I was categorically confused here because the quotient $\mathrm{SL}_2(\mathbb{R})/K$ is a group, so is $\mathbb{H}$ that too? Well, of course not with regular multiplication by the observation of my solution to 1.3b (z^2 need not be in $\mathbb{H}$ when z is). So it is purely set theoretical here.
Surjectivity follows from my solution 1.3a, because for any $i \neq z \in \mathbb{H}$ we can choose a $\gamma \in \mathrm{SL}_2(\mathbb{R})/K$ similar to the construction in equation 2. For $z = i$ we simply choose the identity of course.

Recall injectivity means $\phi(a) = \phi(b) \implies a = b$ for a map ϕ and a, b in the domain of ϕ. Well, then suppose $\gamma \cdot i = \gamma' \cdot i$. This means

$$\begin{aligned} \gamma'^{-1} \cdot (\gamma \cdot i) &= (\gamma'^{-1} \circ \gamma) \cdot i \qquad \text{(by prop 1.2.iii)} \\ &\overset{!}{=} i \end{aligned}$$

so in other words $\gamma'^{-1} \circ \gamma \in K$, hence $[\gamma] = [\gamma']$ in $\mathrm{SL}_2(\mathbb{R})/K$.

1.5

✓

2 Modular Forms for $\mathrm{SL}_2(\mathbb{Z})$

Pensées

- Had to refresh this: 'A **meromorphic** function is a single-valued function that is analytic in all but possibly a discrete subset of its domain, and at those singularities it must go to infinity like a polynomial. A simpler definition states that a meromorphic function is a function $f(z)$ of the form $f(z) = \frac{g(z)}{h(z)}$ for g, h entire functions ($h(z) \neq 0$).' - from Wolfram Mathworld
- Then why do we require meromorphism in the definition of weakly modular?
- I understand the remark that for odd k we only have the zero function being weakly modular, but what is the relevance exactly (or why is that desired)?
- Awaiting my complex analysis self-study, I will for now accept the nice fact that the condition that $\tilde{f}$ can be continued to a meromorphic function on $\mathbb{D}$ is **equivalent** to stating

$$\tilde{f}(q) = \sum_{n=-\infty}^{\infty} a_n q^n, \qquad a_n \in \mathbb{C},\ a_n = 0 \text{ for } n \text{ sufficiently negative(?)} \tag{10}$$

  that is convergent on $\{q \in \mathbb{C} |\ 0 < |q| < \epsilon\}$ for some ϵ. (Whatever sufficiently negative means)
- Right now the definition of a modular form is quite abstract. I need some time to process all components of the definition and why they are relevant: holomorphicity, weakly modular (primarily what the factor of automorphy to the power of k means geometrically for a function).
- Ah I see why we need $k \geq 4$ for the Eisenstein series. First it needs to be even weight by the earlier observation and the proof shows that G_k only converges absolutely for $k \geq 3$.
- It is important that the Eisenstein series is induced by

$$\mathcal{G}_k : \mathcal{L} \longrightarrow \mathbb{C}, \tag{11}$$

$$\Lambda \mapsto \sum_{\omega \in \Lambda \setminus \{0\}} \frac{1}{\omega^k} \tag{12}$$

  with

$$G_k : \mathbb{H} \longrightarrow \mathbb{C}, \tag{13}$$

$$z \mapsto \mathcal{G}_k(\Lambda_z) \tag{14}$$

  because then it is a homogeneous function of weight k by proposition 1.4.
- The expansion of the sum above eq 2.3 is by simply observing

$$\sum_{(n,m) \in \mathbb{Z}^2 \setminus (0,0)} = \sum_{n \neq 0, m = 0} + \sum_{n = 0, m \neq 0} + \sum_{n,m \neq 0} \tag{15}$$

  and then collecting the last two sums as $\sum_{m \neq 0, n \in \mathbb{Z}}$
- A question came to mind: for g, h modular forms, is $f = g/h$ a modular form too? It satisfies the weakly modular part with weight of f being the difference in weights of g and h, but I think it fails the requirement of f being holomorphic on $\mathbb{H}$. Namely it is meromorphic because of potential zeros of h.

- Importance of lemma 2.5: summation order matters
- The first step in the proof of lemma 2.5:

$$\begin{aligned}\sum_{m\neq 0}\sum_{n\in\mathbb{Z}}\frac{1}{(mz+n)^2} &= \sum_{m\neq 0}\sum_{n\in\mathbb{Z}}\frac{1}{(mz+n)^2} - \overbrace{\sum_{m\neq 0}\sum_{n\in\mathbb{Z}}\Big(\frac{1}{mz+n}-\frac{1}{mz+n+1}\Big)}^{=0}\\ &= \sum_{m\neq 0}\sum_{n\in\mathbb{Z}}\Big(\frac{mz+n+1}{(mz+n)^2(mz+n+1)} - \frac{(mz+n)(mz+n+1)}{(mz+n)^2(mz+n+1)} + \frac{(mz+n)^2}{(mz+n)^2(mz+n+1)}\Big)\\ &= \sum_{m\neq 0}\sum_{n\in\mathbb{Z}}\frac{1}{(mz+n)^2(mz+n+1)}\end{aligned}$$

- Adding some steps between the first and final expression of proof 2.4:

$$\begin{aligned}&2\zeta(2)z^{-2} + \sum_{m\neq 0}\sum_{n\in\mathbb{Z}}\frac{1}{(nz-m)^2}\\ &= 2\sum_{j=1}^{\infty}\frac{1}{(jz)^2} + \sum_{n,m\in\mathbb{Z}}\frac{1}{(nz-m)^2} - \sum_{n\in\mathbb{Z}}\frac{1}{(nz)^2}\\ &= 2\sum_{j=1}^{\infty}\frac{1}{(jz)^2} + \sum_{n\neq 0}\sum_{m\in\mathbb{Z}}\frac{1}{(nz-m)^2} - \sum_{n\in\mathbb{Z}}\frac{1}{(nz)^2} + \sum_{m\in\mathbb{Z}}\frac{1}{m^2}\\ &= 2\zeta(2) + \sum_{n\neq 0}\sum_{m\in\mathbb{Z}}\frac{1}{(nz-m)^2}\end{aligned}$$

  Also, this fancy step is necessary

$$z^{-2}G_2(-1/z) + \frac{2\pi i}{z} = 2\zeta(2) + \sum_{n\in\mathbb{Z}}\sum_{m\neq 0}\frac{1}{(mz+n)^2(mz+n+1)}. \tag{16}$$

  because need to show absolute convergence in order to match the exchanged summation order in the definition of G_2. Now using this expression we can expand again as

$$\begin{aligned}&\sum_{n\in\mathbb{Z}}\sum_{m\neq 0}\frac{1}{(mz+n)^2(mz+n+1)} = \sum_{m\neq 0}\sum_{n\in\mathbb{Z}}\frac{1}{(mz+n)^2(mz+n+1)}\\ &= \sum_{m\neq 0}\sum_{n\in\mathbb{Z}}\Big(\frac{1}{(mz+n)^2} + \frac{1}{mz+n} - \frac{1}{mz+n+1}\Big)\\ &= \sum_{m\neq 0}\sum_{n\in\mathbb{Z}}\frac{1}{(mz+n)^2}\end{aligned}$$

- Initially I thought how on earth is the function Δ so strangely defined, but then I thought it is very reminiscent of the discriminant of an elliptic curve which I encountered in my thesis. Also the notation Δ is the same, so curious to see whether it indeed is that.
- I am not immediately familiar with the assertion that when the $\mathbb{C}$-vector space of weight 12 modular forms is one-dimensional and two forms have the same Fourier coefficient for q

that they must be equal. I get that the one-dimensionality implies they are scaled versions of each other, but why does one Fourier coefficient fix equality? Ah this is actually quite trivial. One-dimensionality means $\Delta = c\eta^{24}$ for some $c \in \mathbb{C}$. Then Fourier expansions are $\Delta = \sum_{n\geq 0} a_n q^n$, $\eta^{24} = \sum_{n\geq 0} b_n q^n$ so we get $a_n = cb_n$ and using $a_0 = 1 \overset{!}{=} cb_0 = c$ the equality follows.

- The proof of thm 2.9 is incomplete, eventhough it is quite trivial let me add: Using the valence formula and the fact that for holomorphic functions on $\mathbb{H} \cup \{\infty\}$ all orders are nonnegative integers, we have $k/12 = 1/3$ (for E_4) and $1/2$ (for E_6). The only way this is possible is if $\text{ord}_\rho E_4 = 1$ and $\text{ord}_i E_6 = 1$ and all other orders vanish.
- Corollary 2.10 implicitly answers an important fact: multiplication of two modular forms is another modular form (division is not, by my comment earlier) so does it form a ring? I think yes because $f = 1$ is a modular form which provides the identity and all other axioms seem satisfied. The function $0 : \mathbb{H} \longrightarrow \mathbb{C}$, $z \mapsto 0$ should be the additive identity. One subtlety is of course the weight.
- Loving the simple yet elegant proof of thm 2.12! Because indeed corollary 2.13 allows us to prove two modular forms are identical by comparing finitely many coefficients.

2.1

To be completely honest I did not know the definition of a group action. It does make intuitive sense (Wolfram MathWorld): A group action of a group G on a set X is a map $\phi : G \times X \longrightarrow X$ such that for all $x \in X$

1. $\phi(e, x) = x$ for e the identity in G
2. $\phi(g, \phi(h, x)) = \phi(gh, x)$ for all $g, h \in G$

So we simply check these two requirements.

1. $(f|_k \mathbf{1})(z) = (0z + 1)^{-k} f(\mathbf{1} \cdot z) = f(z) \longrightarrow$ ✓
2. $$\begin{aligned}(f|_k\gamma)(z) &= (cz + d)^{-k} f(\gamma \cdot z) =: g(z) \implies \\ (g|_k\gamma')(z) &= (c'z + d')^{-k} g(\gamma' \cdot z) \\ &= (c'z + d')^{-k}(c(\gamma' z) + d)^{-k} f(\gamma(\gamma' z)) \\ &= (c(a'z + b') + d(c'z + d'))^{-k} f((\gamma\gamma')z) = (*)\end{aligned}$$

   On the other hand, $\gamma\gamma' = \begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix} = \begin{pmatrix} aa' + bc' & ab' + bd' \\ a'c + c'd & b'c + dd' \end{pmatrix}$ so

   $$(f|_k(\gamma\gamma'))(z) = ((a'c + c'd)z + bc' + dd')^{-k} f((\gamma\gamma')z) = (*) \longrightarrow \checkmark$$

2.2

a. I mean this is quite by definition, because writing out $\sigma_t(m)\sigma_t(n)$ I get

$$\begin{aligned}&(1^t + d_{m_1}^t + d_{m_2}^t + \dots m^t)(1^t + d_{n_1}^t + d_{n_2}^t + \dots n^t)\\&= 1^{2t} + 1^t\sigma_t(n) + d_{m_1}^t\sigma_t(n) + \dots d_{m_i}^t\sigma_t(n) + \dots m^t\sigma_t(n) = \sigma_t(nm)\end{aligned}$$

where $d_{n_i}|n$, $d_{m_i}|m$. These are all the divisors of nm (raised to the power of t, as $1^{2t} = 1^t$ of course) since $d|n \implies d|nm$ and $d|m \implies d|nm$ so also $d_{n_i}d_{m_j}|nm$. In fact, all divisors of nm are of this form. Furthermore, because $\gcd(n, m) = 1$ there is also no over counting.

b.The desired expression screams geometric series: $\sum_{k=0}^{N} r^k = \frac{r^{N+1}-1}{r-1}$. Indeed, using the result of part (a.) we see inductively that

$$\sigma_t(n) = \prod_p \sigma_t(p^{e_p})$$

and

$$\begin{aligned}\sigma_t(p^{e_p}) &= \sum_{d|p^{e_p}} d^t = \sum_{j=0}^{e_p} p^{jt}\\&= \frac{p^{e_p+1} - 1}{p^t - 1}\end{aligned}$$

where I used the fact that p is prime, so all divisors of p^{e_p} are powers of p up to e_p. The desired expression follows.

2.3

a. Recall the generating function $\frac{t}{e^t-1} = \sum_{k=0}^{\infty} \frac{B_k}{k!}t^k$. Then we simply first send $t \mapsto 2\pi iz$:

$$\begin{aligned}\frac{t}{e^t - 1} = \frac{2\pi iz}{e^{2\pi iz} - 1} &= 2\pi iz\frac{1}{e^{\pi iz}(e^{\pi iz} - e^{-\pi iz})}\\&= \pi z\frac{1}{e^{\pi iz}\sin(\pi z)}\\&= \pi z\frac{e^{-\pi iz}}{\sin(\pi z)} = \sum_{k=0}^{\infty}\frac{B_k}{k!}(2\pi i)^k z^k.\end{aligned}$$

Likewise for $t \mapsto -2\pi iz$ I get

$$\begin{aligned}\frac{t}{e^t - 1} &= -2\pi iz\frac{1}{e^{-\pi iz}(e^{-\pi iz} - e^{\pi iz})}\\&= \pi z\frac{e^{\pi iz}}{\sin(\pi z)} = \sum_{k=0}^{\infty}\frac{B_k}{k!}(-2\pi i)^k z^k.\end{aligned}$$

Now if we add both terms I get

$$\begin{aligned}\pi z\frac{e^{\pi iz} + e^{-\pi iz}}{\sin(\pi z)} &= 2\sum_{k\geq 0 \text{ even}}\frac{B_k}{k!}(2\pi i)^k z^k\\\iff \pi z\frac{\cos(\pi z)}{\sin(\pi z)} &= \sum_{k\geq 0 \text{ even}}(2\pi i)^k\frac{B_k}{k!}z^k.\end{aligned}$$

b. Equation (A.1) reads $\pi\frac{\cos(\pi z)}{\sin(\pi z)} = \frac{1}{z} + \sum_{n=1}^{\infty}\Big(\frac{1}{z-n} + \frac{1}{z+n}\Big)$. In other words,

$$\begin{aligned}
&\frac{1}{z} + \sum_{n=1}^{\infty}\Big(\frac{1}{n(\frac{z}{n}-1)} + \frac{1}{n(\frac{z}{n}+1)}\Big) \\
&= \frac{1}{z} - \sum_{n=1}^{\infty}\sum_{k=0}^{\infty}\frac{1}{n}\Big((\frac{z}{n})^k - (-\frac{z}{n})^k\Big) \qquad \text{(geometric series)} \\
&= \frac{1}{z} - 2\sum_{\substack{k\geq 1\\ k \text{ odd}}}\sum_{n=1}^{\infty}\frac{1}{n}(\frac{z}{n})^k \\
&= \frac{1}{z} - 2\sum_{k\geq 2 \text{ even}} z^{k-1}\sum_{n=1}^{\infty}\frac{1}{n^k} \\
&= \frac{1}{z} - 2\sum_{k\geq 2 \text{ even}} \zeta(k) z^k
\end{aligned}$$

so multiplying both sides by z I indeed find

$$\frac{\pi z\cos\pi z}{\sin\pi z} = 1 - 2\sum_{\substack{k\geq 2\\ k \text{ even}}}\zeta(k)\, z^k. \tag{17}$$

c. This is straightforwardly writing out:

$$\begin{aligned}
\sum_{k\geq 0 \text{ even}}(2\pi i)^k\frac{B_k}{k!}z^k &= B_0 + \sum_{k\geq 2 \text{ even}}(2\pi i)^k\frac{B_k}{k!}z^k \\
&\overset{!}{=} 1 - 2\sum_{k\geq 2 \text{ even}}\zeta(k)z^k \qquad \forall z \implies \\
-2\zeta(k) &\overset{!}{=} (2\pi i)^k\frac{B_k}{k!} \iff \\
\zeta(k) &= -\frac{(2\pi i)^k}{2\cdot k!}B_k \qquad \forall k\geq 2
\end{aligned}$$

d. This requires looking at the generating function again:

$$\frac{t}{e^t-1} = \sum_{k=0}^{\infty}\frac{B_k}{k!}t^k.$$

I immediately thought we need to consider $t\mapsto -t$ in order to somehow get cancelling terms, as we want to show vanishing of B_k for $k>1$ odd. Upon close inspection I noticed something nice upon $t\mapsto -t$:

$$\begin{aligned}
&\frac{t}{e^t-1} \mapsto \frac{-t}{e^{-t}-1} = \frac{t}{1-e^{-t}} = \frac{te^t}{e^t-1} \implies \\
&\frac{te^t}{e^t-1} - \frac{t}{e^t-1} = t
\end{aligned}$$

Therefore, we must have

$$t \overset{!}{=} \sum_{k=0}^{\infty} \left(\frac{B_k}{k!}(-1)^k t^k - \frac{B_k}{k!} t^k \right)$$
$$\implies 1 \overset{!}{=} -2B_1 \quad \text{and} \quad B_k \overset{!}{=} 0 \, \forall k > 1 \text{ odd}.$$

2.4

a. Doesn't this immediately follow from its relationship with $E_4(z)$ which has a simple zero at $z = \rho = \exp\left(\frac{2\pi i}{3}\right)$ by thm 2.9? (yeah, but this question was probably intended to show a different way)
This I saw immediately geometrically by looking at the fundamental domain depicted in chapter 1.3. Namely, $\exp(2\pi i/3) = \rho$ so

$$\begin{aligned} G_4(-1/\rho) &= G_4(\rho + 1) \qquad \text{(by inspecting the figure (or simple algebra))} \\ &= G_4(\rho) \qquad \text{(modularity)} \\ &\overset{!}{=} \rho^4 G_4(\rho) \qquad \text{(by the hint provided)} \\ &= \rho G_4(\rho) \end{aligned}$$

so as $\rho \neq 0$, we must have $G_4(\rho) = 0$.

b. By the same procedure this one is a piece of cake. $G_6(-1/i) = G_6(i) \overset{!}{=} i^6 G_6(i) = -G_6(i) \implies G_6(i) = 0$.

2.5

I immediately note that $S \in \mathrm{SL}_2(\mathbb{Z})$ is satisfied by proposition 2.4 and captured by the transformation law written in the problem here. Also, T simply reads $E_2(z+1) = E_2(z)$ so that also checks out. The fact that $\mathrm{SL}_2(\mathbb{Z})$ is generated by S, T means that any element of the group is a product of powers of those matrices. Does it suffice to check ST and TS or something?
The ingredients are there, but I can't seem to cook the recipe yet. I will get back to this next week! (22-09-2026:) The idea I have is some kind of induction. Namely, every $\gamma \in \mathrm{SL}_2(\mathbb{Z})$ is a product of powers of S, T. So given the proposed equation holds for S, T, then we can write $E_2(\gamma z) = E_2(Sw)$, with $w = S^{-1}\gamma z$, or $E_2(\gamma z) = E_2(Tv)$, with $v = T^{-1}\gamma z$ depending on whether there is a T or S on the left when γ is written as a product of them. Until eventually we simply end up with Sz or Tz in the argument. To that end I think it suffices to check 'associativity' of T^2, TS, ST and S^2 in the following sense.
Proposition 2.4 reads

$$E_2(S \cdot z) = z^2 E_2(z) - \frac{z}{4\pi i} \tag{18}$$

So we must have

$$\begin{aligned} E_2((S^2)z) &= E_2(Sw) \qquad (w = S\cdot z = -\frac{1}{z}) \\ &= w^2 E_2(w) - \frac{w}{4\pi i} \qquad \text{(by eq. 18)} \\ &= \frac{1}{z^2}E_2(-\frac{1}{z}) + \frac{1}{4\pi i z} \\ &= \frac{1}{z^2}\Big(z^2 E_2(z) - \frac{z}{4\pi i}\Big) + \frac{1}{4\pi i z} \qquad \text{(by eq. 18)} \\ &= E_2(z) \end{aligned}$$

And on the other hand $S^2 = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$ so the problem's proposed equation immediately reads $(0z-1)^{-2}E_2(S^2\cdot z)(= E_2(z)) \overbrace{=}^{\text{RHS}} E_2(z) - \frac{1}{4\pi i}\frac{0}{0z-1} = E_2(z)$ yielding the same result as above.

Likewise it is straightforward to show that TS, ST, T^2 also satisfy this associative property, but I remain unsure whether this is sufficient so I am a bit hesitant to write this out properly. Will get back to it later. I am also considering the option computing this associative property for $S^{-1}\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $T^{-1}\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with my proposed procedure. That at least 'feels' more like the right thing to do, although they are the same in spirit. The latter works as follows: $S^{-1}\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} -c & -d \\ a & b \end{pmatrix}$ and letting $w = S^{-1}\gamma z$, I get

$$\begin{aligned} E_2(\gamma z) &= E_2(Sw) \\ &= w^2 E_2(w) - \frac{w}{4\pi i} \qquad \text{(by prop. 2.4)} \\ &= \Big(\frac{-cz-d}{az+b}\Big)^2\Big((az+b)^2 E_2(z) - \frac{1}{4\pi i}\frac{a}{az+b}\Big) - -\frac{1}{4\pi i}\frac{cz+d}{az+b} \quad \text{(by induction on proposed formula)} \\ &= (cz+d)^2\Big(E_2(z) - \frac{1}{4\pi i}\frac{c}{cz+d}\Big) \qquad \text{(using } ad-bc=1) \end{aligned}$$

Likewise, $T^{-1}\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a-c & b-d \\ c & d \end{pmatrix}$ so

$$\begin{aligned} E_2(\gamma z) = E_2(Tw') \qquad (w = T^{-1}\gamma z = \frac{(a-c)z+b-d}{cz+d}) \\ = E_2(w+1) = E_2(w) = (cz+d)^2(E_2(z) - \frac{1}{4\pi i}\frac{c}{cz+d}) \end{aligned}$$

so both agree with the proposed equation and like this we can strip away T or S iteratively from γ (which is a product of powers of T, S) until we are left with z in the argument of E_2.

2.6

a. This is relatively straightforward: $E_k(z) = \frac{(k-1)!}{2(2\pi i)^k} G_k(z)$ so $E_2(z) = \frac{1}{2(2\pi i)^2} G_2(z)$ thus

$$\begin{aligned}
f(\gamma z) &= G_2(\gamma z) - \frac{\pi}{\mathrm{Im}(\gamma z)} = 2(2\pi i)^2 E_2(\gamma z) - \frac{\pi}{\mathrm{Im}(\gamma z)} \\
&= 2(2\pi i)^2 \Big[j(\gamma, z)^2 \big(E_2(z) - \frac{1}{4\pi i} \frac{c}{cz+d} \big) \Big] - \frac{\pi}{\mathrm{Im}(\gamma z)} \qquad \text{(exercise 2.5)} \\
&= j(\gamma, z)^2 G_2(z) - j(\gamma, z)^2 2\pi i \frac{c}{cz+d} - \frac{\pi}{\mathrm{Im}(\gamma z)} \\
&= j(\gamma, z)^2 \Big[G_2(z) - 2\pi i \frac{c}{cz+d} \Big] - \frac{\pi}{\mathrm{Im}(z)} |cz+d|^2 \\
&= j(\gamma, z)^2 \Big[G_2(z) - 2\pi i \frac{c}{cz+d} - \frac{\pi(c\overline{z}+d)}{\mathrm{Im}(z)(cz+d)} \Big] \\
&= j(\gamma, z)^2 \Big[G_2(z) - \pi \frac{c(z-\overline{z})}{(cz+d)\mathrm{Im}(z)} - \frac{\pi(c\overline{z}+d)}{\mathrm{Im}(z)(cz+d)} \Big] \qquad (\mathrm{Im}(z) = \frac{1}{2i}(z-\overline{z})) \\
&= j(\gamma, z)^2 \Big[G_2(z) - \frac{\pi(cz+d)}{(cz+d)\mathrm{Im}(z)} \Big] = j(\gamma, z)^2 f(z).
\end{aligned}$$

b. Yes, $\frac{1}{\mathrm{Im}(z)}$ is holomorphic on $\mathbb{H} \cup \{\infty\}$ and part a. proves weight two weak modularity.

2.7

Recall $\Delta = \frac{(240E_4)^3 - (-504E_6)^2}{1728}$. Here I initially thought why is there a minus in front of 504, but it is to normalise the constant term to one:

$$E_4 = -\frac{B_4}{8} + \sum_{n=1}^{\infty} \sigma_3(n) q^n, \tag{19}$$

$$-\frac{B_4}{8} = --\frac{1}{8 \cdot 30} = \frac{1}{240} \tag{20}$$

$$E_6 = -\frac{B_6}{12} + \sum_{n=1}^{\infty} \sigma_5(n) q^n, \tag{21}$$

$$-\frac{B_6}{12} = -\frac{1}{504}. \tag{22}$$

My immediate idea was to expand the terms in Δ and seeing whether each coefficient is divisible by 1728.

$$\begin{aligned}
\Delta &= \frac{(1 + 240 \sum_{n=1}^{\infty} \sigma_3(n) q^n)^3 - (1 - 504 \sum_{n=1}^{\infty} \sigma_5(n) q^n)^2}{1728} \\
&= \frac{1}{1728} \Big[1 + 3 \cdot 240 \sum_{n=1}^{\infty} \sigma_3(n) q^n + 3 \big(240 \sum_{n=1}^{\infty} \sigma_3(n) q^n \big)^2 + \big(240 \sum_{n=1}^{\infty} \sigma_3(n) q^n \big)^3 \\
&- \Big(1 - 2 \cdot 504 \sum_{n=1}^{\infty} \sigma_5(n) q^n + \big(504 \sum_{n=1}^{\infty} \sigma_5(n) q^n \big)^2 \Big) \Big]
\end{aligned}$$

the 1's cancel and every coefficient is divisible by 1728 except

$$\sum_{n=1}^{\infty} \big(3 \cdot 240 \sigma_3(n) + 2 \cdot 504 \sigma_5(n) \big) q^n \tag{23}$$

which must therefore somehow be divisible by 1728 too... Ah, perhaps ideas from exercise 8,9 might be useful? Hmm, cannot have a relation with σ_3 and σ_5 because we would need $E_6 = cE_4E_2$, but E_2 is not a modular form and this would involve $\sigma_1(n)$ too.
Let's try something else. $1728 = 2^6 \cdot 3^3$, $3 \cdot 240 = 2^4 \cdot 3^2 \cdot 5$ and $2 \cdot 504 = 2^4 \cdot 3^2 \cdot 7$ so dividing out common terms,

$$\frac{1}{1728}\Big(\sum_{n=1}^{\infty}\big(3\cdot 240\sigma_3(n) + 2\cdot 504\sigma_5(n)\big)q^n\Big) = \frac{1}{12}\Big(\sum_{n=1}^{\infty}\big(5\sigma_3(n)+7\sigma_5(n)\big)q^n\Big)$$
$$\implies \text{must prove: } \frac{5\sigma_3(n)+7\sigma_5(n)}{12} \in \mathbb{Z}\ \forall n$$

Let's first see if this is true for prime numbers $n = p$. Then the only divisors are, by definition, 1 and p so $\sigma_3(p) = 1 + p^3$ and $\sigma_5(p) = 1 + p^5$ yielding

$$\frac{5p^3+7p^5}{12} = \frac{(5+7p^2)p^3}{12}, \checkmark \text{ for } p = 2, 3$$

Next for any $p > 3$ we must apparently have that $12|(5+7p^2)$ (as $12 = 2^2\cdot 3$). To prove this, a fact I encountered during a group theory course came to min: $a|x$ and $b|x \implies \text{lcm}(a,b)|x$ so if we prove $3|(5+7p^2)$ and $4|(5+7p^2)$ we are done. Well

$$5 + 7p^2 \bmod 3 = \overline{2} + \overline{p^2} \overset{!}{=} \overline{0}$$
$$\implies \overline{p^2} = \overline{p}^2 \overset{!}{=} \overline{1}$$
$$\iff \overline{p} = \overline{1} \text{ or } \overline{p} = \overline{2}$$

which is true for any prime number $p > 3$. Likewise

$$5 + 7p^2 \bmod 4 = \overline{1} + \overline{3}\overline{p^2} \overset{!}{=} \overline{0}$$
$$\implies \overline{p^2} = \overline{p}^2 \overset{!}{=} \overline{1}$$
$$\iff \overline{p} = \overline{1} \text{ or } \overline{p} = \overline{3}$$

which is again true for any prime number $p > 3$ so $12|(5+7p^2)\ \forall p > 3$, thus $\frac{(5+7p^2)p^3}{12} \in \mathbb{Z}$ for every prime number.
Now we can probably use the result of problem 2.2 somehow...The generalisation is not obvious to me at the moment. Will get back to it after continuing with 11-15

2.8

$|M_8| = 1 \implies f = cE_8 = c\Big(-\frac{B_8}{16} + \sum_{n=1}^{\infty}\sigma_7(n)q^n\Big)$ for some $c \in \mathbb{C}$ and $f \in M_8$. We also know $E_4^2 \in M_8$ where $E_4 = \frac{1}{240} + \sum_{n=1}^{\infty}\sigma_3(n)q^n$ so

$$E_4^2 = \big(\frac{1}{240}\big)^2 + \frac{1}{120}\sum_{n=1}^{\infty}\sigma_3(n)q^n + \Big(\sum_{n=1}^{\infty}\sigma_3(n)q^n\Big)^2$$
$$\overset{!}{=} cE_8 \overset{\text{Cor. 2.13}}{\implies} \frac{c}{480} \overset{!}{=} \frac{1}{240^2} \iff c = \frac{1}{120}.$$

In other words,

$$\frac{1}{120}\sum_{n=1}^{\infty}\sigma_7(n)q^n = \frac{1}{120}\sum_{n=1}^{\infty}\sigma_3(n)q^n + \Big(\sum_{n=1}^{\infty}\sigma_3(n)q^n\Big)^2 \tag{24}$$

and requiring this to hold for all q we find

$$\sigma_7(n) = \sigma_3(n) + 120\sum_{j=1}^{n-1}\sigma_3(j)\sigma_3(n-j). \tag{25}$$

2.9

a. Like the previous problem, $|M_{10}| = 1$ so $E_{10} = cE_4E_6$ for some $c \in \mathbb{C}$ yielding $-\frac{B_{10}}{20} \overset{!}{=} c\frac{-B_6}{12}\cdot\frac{-B_4}{8}$ by corollary 2.13 again $\implies c = \frac{5040}{11}$.

$$E_{10} = cE_4E_6 = c\Big[\big(\frac{1}{240} + \sum_{n=1}^{\infty}\sigma_3(n)q^n\big)\big(-\frac{1}{504} + \sum_{n=1}^{\infty}\sigma_5(n)q^n\big)\Big] \implies$$

$$\sum_{n=1}^{\infty}\sigma_9(n)q^n = c\frac{1}{240}\sum_{n=1}^{\infty}\sigma_5(n)q^n - c\frac{1}{504}\sum_{n=1}^{\infty}\sigma_3(n)q^n + c\sum_{n=1}^{\infty}\sum_{j=1}^{n-1}\sigma_3(j)\sigma_5(n-j)q^n \implies$$

$$\sigma_9(n) = \frac{21}{11}\sigma_5(n) - \frac{10}{11}\sigma_3(n) + \frac{5040}{11}\sum_{j=1}^{n-1}\sigma_3(j)\sigma_5(n-j).$$

b. This follows the same line: $E_{14} = cE_4E_{10} \implies -\frac{1}{24} = c\frac{1}{240}\cdot\frac{-5}{20\cdot 66} \implies c = 2640$ obtaining the relation

$$\sigma_{13}(n) = c\frac{-B_4}{8}\sigma_9(n) + c\frac{B_{10}}{20}\sigma_3(n) + c\sum_{j=1}^{n-1}\sigma_3(j)\sigma_9(n-j) \tag{26}$$

Likewise $E_{14} = \tilde{c}E_6E_8 \implies \tilde{c} = 10080$ yielding

$$\sigma_{13}(n) = \tilde{c}\frac{-B_6}{12}\sigma_7(n) + \tilde{c}\frac{-B_8}{16}\sigma_5(n) + \tilde{c}\sum_{j=1}^{n-1}\sigma_5(j)\sigma_7(n-j) \tag{27}$$

c. Really same Spiel.... $E_{14} = cE_4^2E_6$ finding $c = \frac{12B_{14}}{28B_6}\big(\frac{8}{B_4}\big)^2$ etc. etc., tedious but straightforward.

2.10

a. I think this statement is trivial by using thm 1.5: that every point in $\mathbb{H}$ is $\mathrm{SL}_2(\mathbb{Z})$ equivalent to a point in the fundamental domain. Because the lowest (right) corner is at $\rho + 1 = \frac{1+i\sqrt{3}}{2}$ which has imaginary part $C = \sqrt{3}/2$, qed.

b. A weight zero modular form f satisfies $f(\gamma z) = f(z)$, $\forall z \in \mathbb{H}$ and $\forall\gamma \in \mathrm{SL}_2(\mathbb{Z})$. By part a. we can thus assume $\mathrm{Im}(z) \geq \sqrt{3}/2$ and a cusp form f has q-expansion $f = \sum_{n=1}^{\infty}a_nq^n$ so

$$|f| = \sum_{n=1}^{\infty}|a_n||q^n|,$$

$$|q^n| = |\exp(2\pi i(\mathrm{Re}(z) + i\mathrm{Im}(z)))n|$$

$$= \exp(-2\pi n\mathrm{Im}(z)) \leq \exp\Big(-\sqrt{3}\pi n\Big)$$

so indeed $|f|$ attains a maximum on $\mathbb{H}$, precisely where the bound is saturated.

c. By thm A.4 the cusp form f must be constant on $\mathbb{H}$. I don't immediately see how I must now conclude that every cusp form must be zero and what remains is constant modular forms of weight zero... I guess that we can write for every $\tilde{f} \in M_0$ that $\tilde{f} = c + f$, for $c \in \mathbb{C}$ and $f \in S_0$ and I think the weight zero modularity property together with the fact that f is constant ought to force $f = 0$... Ah yes of course, because cusp forms satisfy $f(\infty) = 0$ which is only possible when $f = 0$ altogether (as f =const. by b.).

2.11

a. $\Delta = \lambda E_4^3 + \mu E_{12}$ can be solved for λ and μ using corollary 2.13 and the first few terms of the expansion of Δ; $a_0 = 0$, $a_1 = 1$. This quickly gives the system of equations

$$\lambda\big(-\frac{B_4}{8}\big) - \mu\frac{B_1 2}{24} = 0,$$
$$3\lambda\big(-\frac{B_4}{8}\big)^2\sigma_3(1) + \mu\sigma_{11}(1) = 1.$$

Noting that $\sigma_t(1) = 1\,\forall t$ and $B_4 = -\frac{1}{30}$, $B_{12} = -\frac{691}{2730}$ we can let Mathematica solve this and I found $\lambda = 22112$, $\mu = -\frac{91}{600}$.

b. As $691|22112$, it immediately follows that $\Delta \equiv \mu E_{12} \bmod 691$. I.e. $600\Delta \equiv -91E_{12} \bmod 691 \equiv 600E_{12} \bmod 691$, hence $\tau(n) \equiv \sigma_{11}(n) \bmod 691$.

2.12

a. Let us write $g(z) = f'(z) + 4\pi ikE_2(z)f(z)$. We need to show $g \in M_{k+2}$ when $f \in M_k$. To that end I will first show weak modularity.

$$\begin{aligned} g(\gamma z) &= f'(\gamma z) + 4\pi ikE_2(\gamma z)f(\gamma z) \\ &= f'(\gamma z) + 4\pi ik(cz+d)^2\Big(E_2(z) - \frac{1}{4\pi i}\frac{c}{cz+d}\Big)(cz+d)^k f(z) \quad \text{(by prop. 2.4 and } f \in M_k) \\ &= f'(\gamma z) + (cz+d)^{k+2}4\pi ikE_2(z)f(z) - (cz+d)^{k+1}ckf(z). \end{aligned}$$

So if we can show $f'(\gamma z) = (cz+d)^{k+2}f'(z) + (cz+d)^{k+1}ckf(z)$ this part is done. We need to be careful in notation here: $f'(w)$ means $\frac{\partial f}{\partial w}$ and $(f(w)' = \frac{\partial f}{\partial w}\frac{\mathrm{d}w}{\mathrm{d}z}$

$$\Big(f(\gamma z)\Big)' = f'(\gamma z)(\gamma z)' = \frac{1}{(cz+d)^2}f'(\gamma z)$$

but also

$$\begin{aligned} \Big(f(\gamma z)\Big)' &= \Big((cz+d)^k f(z)\Big)' \\ &= ck(cz+d)^{k-1}f(z) + (cz+d)^k f'(z) \implies \\ f'(\gamma z) &= ck(cz+d)^{k+1}f(z) + (cz+d)^{k+2}f'(z), \; qed. \end{aligned}$$

!It remains to argue that $g(z)$ is holomorphic on $\mathbb{H} \cup \{\infty\}$!. Of course $f'(z)$ is holomorphic, because by assumption $f(z)$ is too.

b. I think we can show closure upon differentiation for monomials E_2, E_4, E_6 and qed, because then it follows for any polynomial by the product rule. In the previous part we found $g(z) = f'(z) + 4\pi i k E_2(z) f(z)$ and that for any $f \in M_k$ we have $g \in M_{k+2}$. Letting $f = E_2$, I obtain $g = E_2' + 8\pi i E_2^2$, but also $\dim(M_4) = 1$ so $g = \alpha E_4$ for some $\alpha \in \mathbb{C}$. Hence, $E_2' = \alpha E_4 - 8\pi i E_2^2 \in \mathbb{C}[E_2, E_4, E_6]$. We play the same game for E_4, E_6: $f = E_4 \implies E_4' = \alpha E_6 - 16\pi i E_2 E_4$ (not necessarily the same α as before) and for $f = E_6 \implies E_6' = \alpha E_8 - 24\pi i E_2 E_6$, but $E_8 = \beta E_4^2$ for some $\beta \in \mathbb{C}$ as $\dim(M_8) = 1$ as well. I think this is the idea, but it feels a bit shaky still as E_2 is not a modular form. However, the transformation rule of part a. still holds for E_2 too.

2.13

a. This seems rather obvious, because modular functions are of the form f/g for $f, g \in M_k$ and $g \neq 0$. Then $1 \in F$ (take $f = 1 = g$) and the inverse is always given (for non-zero elements) by g/h.

b. Do they mean the j-function here?? I'm super confused because I am only familiar with 'numbers' being transcendental over some field (like π over $\mathbb{Q}$) and since $\mathbb{C}$ is algebraically closed I'm a bit puzzled. But then I looked it up and functions can also be transcendental by the same requirement: If $\nexists f \in \mathbb{C}[x]$ such that $f(j) = 0$ (the zero function), then j is a transcendental function over $\mathbb{C}$. But I know that for $K \subsetneq \overline{K} \subsetneq L$, with $\overline{K}$ the algebraic closure of K, we have that for all elements $\alpha \in L$ not contained in $\overline{K}$ that α is transcendental over K. Does it then suffice to remark that $j \notin \mathbb{C}$?? Definitely not, because then you could say that any polynomial is a transcendental function... I need to figure out how this translates to transcendental functions.

2.14

a. We know $j = \frac{(240E_4)^3}{\Delta}$, with $\Delta = \frac{(240E_4)^3 - (-504E_6)^2}{1728}$, and $E_k \propto G_k$. In problem 4 we saw that $G_4(\rho) = 0$ and $G_6(i) = 0$, so $E_4(\rho) = 0 = E_6(i)$. Using this, it immediately follows that $j(i) = 1728$ and $j(\rho) = 0$.

b. $\mathrm{Re}(z) = 0$ is easy, because then $E_4(z)$ and $E_6(z)$ are real and therefore $j(z) \in \mathbb{R}$.[3] Also the left and right boundary edges of $\mathcal{D}$ are quite straightforward because $q(\mathrm{Re}(z) = \pm\frac{1}{2}) = \exp\big(2\pi i(\pm\frac{1}{2} + bi)\big) = \exp(\pm\pi i)\exp(-2\pi b) \in \mathbb{R}$ (so the entire q-expansion of j is real too).
It remains to show the arc part of the boundary of $\mathcal{D}$. To that end we can exploit $j(\gamma z) = j(z)$ for all $j \in \mathrm{SL}_2(\mathbb{Z})$. Namely on this part of the boundary $|z| = 1$ so given $z \sim -\frac{1}{z} = -\frac{\overline{z}}{|z|^2} = -\overline{z}$ we must have

$$\begin{aligned} j(z) &= j(-\overline{z}) \\ &= \overline{j(z)} \implies j(z) \in \mathbb{R} \end{aligned}$$

where the last step can be seen by $\overline{q} = \exp(-2\pi i \overline{z}) = q(-\overline{z})$.

c. Well-definedness follows immediately by $j(\gamma z) = j(z)\ \forall \gamma \in \mathrm{SL}_2(\mathbb{Z})$. The bijective part of this question has been bugging me for days... (!)

[3] This can be seen by the q-expansion and recalling $q = \exp(2\pi i z)$.

d. Requiring $j(z) = \overline{j(z)} = j(-\overline{z}) \implies$ there must be a $\gamma \in \mathrm{SL}_2(\mathbb{Z})$ such that $z = \gamma \cdot -\overline{z}$. Since we take all $z \in \mathcal{D}$ (representative of domain of the map $\overline{j}$), we have 4 cases.

1. $\gamma = S$: $z = \frac{1}{\overline{z}} \implies |z| = 1$ so this is the circular arc of $\mathcal{D}$.
2. $\gamma = T$: $z = -\overline{z} + 1 \iff a + bi = 1 - a + bi \implies a = \frac{1}{2}$. So this is the right boundary of $\mathcal{D}$.
3. $\gamma = T^{-1}$: $z = -\overline{z} - 1 \iff a + bi = -1 - a + bi \implies a = -\frac{1}{2}$. So this is the left boundary of $\mathcal{D}$.
4. $\gamma = \mathbf{1}$: $z = -\overline{z}$ is satisfied when $\mathrm{Re}(z) = 0$.

2.15

a. First I was confused 'what about M_2, M_0?', but $|M_2| = 0$ of course (and $4a + 6b = 0$ has no solutions) and $|M_0| = 1$ (corresponding with the unique solution $(a, b) = (0, 0)$ of $4a + 6b = 0$). I secretly already use the result of part b that E_4 and E_6 are independent so the different solutions to the equation correspond with the dimension of M_k. Next, I thought what's special about 2? But then I saw that $k = 0, 4, 6, 8, 10$ all have a single solution for $4a + 6b = k$ and $k = 12$ has two, but $k = 2$ has none and I realised that for every multiple of 12 you get an extra solution to the equation. That is because suppose $4a+6b = x$ is a solution, then $4a'+6b' = x+12$ has an additional solution: $(a', b') = (a + 3, b)$ or $(a', b') = (a, b + 2)$. The solution to the first equation gets shifted to a new solution and there is an additional one. Of course this would mean that every solution would provide a new solution when $k \mapsto k + 12$, but this would overcount in for example $4a + 6b = 12$: it has solutions $(3, 0)$, $(0, 2)$ and $4a' + 6b' = 24$ would then give $(6, 0)$, $(3, 2)$ and $(3, 2)$, $(0, 4)$ as new solutions which overcounts the middle solution $(3, 2)$.

b. I had to look up algebraic independence from Wikipedia: A subset $S \subseteq L$ of a field L is algebraically independent over $K \subseteq L$ if there exists no non-zero polynomial $P(x_1, \ldots, x_n) \in K[x_1, \ldots, x_n]$ such that $P(s_1, \ldots, s_2) = 0$ for all $s_1, \ldots, s_n \in S$.
For example, the sets $\{\sqrt{\pi}\}$ and $\{2\pi + 1\}$ are algebraically independent over $\mathbb{Q}$ because they are transcendental over $\mathbb{Q}$, but the set $\{\sqrt{\pi}, 2\pi+1\}$ is not. That is because $P(x, y) = 2x^2-y+1 \in \mathbb{Q}[x, y]$ has $P(\sqrt{\pi}, 2\pi + 1) = 0$.

Now back to the question, suppose there were a polynomial $P(x, y) \in \mathbb{C}[x, y]$ such that $P(E_4, E_6) = 0\, \forall z \in \mathbb{H}$. Then I think the issue comes from the transformation law that upon $z \mapsto -\frac{1}{z}$:

$$E_4(z) \mapsto E_4(-\frac{1}{z}) = z^4 E_4(z),$$
$$E_6(z) \mapsto E_6(-\frac{1}{z}) = z^6 E_6(z).$$

Given that they don't transform the same way and $P(E_4, E_6) = 0\, \forall z \in \mathbb{H}$, we must have that $P(x, y)$ is homogeneous. I mean it must be built from the monomials from part a.: $E_4^a E_6^b$ such that $4a + 6b = \deg(P) \in \mathbb{Z}_{\geq 0}$. Now I must somehow conclude that this is only possible when $P = 0$ altogether.

Closing comments

The last five problems felt significantly harder than the previous ten. I am still not satisfied with some, but for now I will continue with chapter three and hope to solve them later or ask help from someone more knowledgeable than I am. My solutions to 7c, 12a, 13b, 14c and 15 are incomplete.